Modeling and Analysis of Soft Robotic Fingers using the Fin Ray Effect

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Abstract

Soft grasping of random objects in unstructured environments has been a research topic of predilection both in academia and in the industry because of its complexity but great practical relevance. However, accurate modeling of soft hands and fingers has proven a difficult challenge to tackle. Focusing on this issue, this paper presents a detailed mathematical modeling and performance analysis of parallel grippers equipped with soft fingers taking advantage of the Fin Ray Effect. The Fin Ray Effect (FRE), based on biomimetic principles, is most commonly found in the design of grasping soft fingers but despite their popularity, finding a convenient model to assess the grasp capabilities of these fingers is challenging. This paper aims at solving this issue by providing an analytic tool to better understand and ultimately design this type of soft fingers. First, a kinetostatic model of a general multi-crossbeam finger is established. This model will allow for a fast yet accurate estimation of the contact forces generated when the fingers grasp an arbitrarily shaped object. The obtained mathematical model will be subsequently validated by numerically to ensure the estimations of the overall grasp strength and individual contact forces are indeed accurate. Physical experiments conducted with 3D-printed fingers of the most common architecture of Fin Ray Effect fingers will also be presented and shown to support the proposed model. Finally, the impact of the relative stiffness between different areas of the fingers will be evaluated to provide insight into further refinement and optimization of these fingers.

Keywords

soft robotics, compliant mechanisms, grasping, Fin Ray Effect, industrial grippers

1 Introduction

Adaptive grasping is not only an attractive topic in the design of compliant and underactuated robotic hands as discussed in Birglen et al. (2008), but also one of the main areas of application of soft robotics, see Hughes et al. (2016); Iida and Laschi (2011) for example. The passive compliance and adaptability of soft robotic systems are often argued to offer unrivaled performance to handle deformable, fragile, and even time-varying objects Wang et al. (2015). Such soft mechanisms constitute a recent branch of robotics closely related to the more common compliant mechanisms that refers to linkages relying on structural deformation to transmit forces and motions. Soft robotics on the other hand also includes possible plastic deformation and is dynamically configuration-dependent in terms of softness as discussed in Chen et al. (2017). With respect to these properties, soft robotics goes beyond the traditional scope of applications of compliant mechanisms. Drawing from the substantial literature available about these compliant mechanisms, this paper discuss how to use the well-known lumped compliance modeling technique of linkages and associated multibody analysis to compute an accurate model for a very popular type of soft fingers using what is referred to as the Fin Ray Effect (FRE) presented by Bannasch and Kniese (2012) and Festo Gmbh (2014).

The FRE is based on biomechanical principles observed in the fins of fishes and its replication in grasping tools allows for a structural, soft and compliant envelopment of objects without any embedded actuation: the FRE fingers

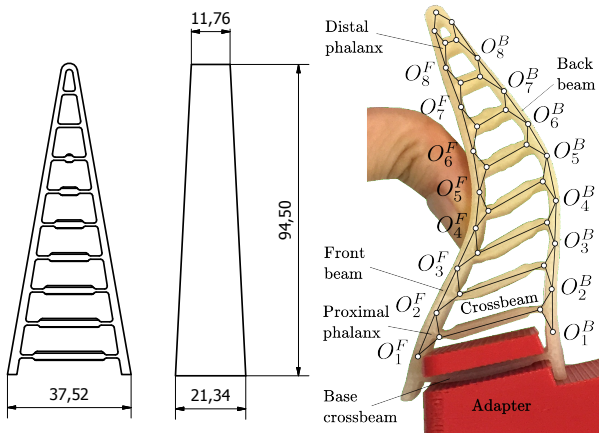


Figure 1. A typical FRE finger: (a) dimensional drawing (mm); (b) 3D-printed prototype plotted with a schematic representation

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are simply pushed against the objects to be seized and they deform in reaction to the resulting contacts. Pioneered by the automation company Festo and available as a commercial product since the end of 2016, FRE fingers permit to transform any industrial gripper into a soft adaptive hand at a very modest cost.

FRE fingers are also simplistic to use and classic on/off pneumatic grippers are perfectly well suited for driving them. Practically, FRE fingers are soft and triangular with crossbeams connecting the front and the back of the fingers. These different parts can bend and deform to conform around objects of various sizes and properties. The dimensions of a size 80 commercial Festo DAHS finger (their market name) as listed in Festo Gmbh (2017) are illustrated in Fig. 1(a). As can be seen in this figure, there are nine crossbeams connecting the front and back beams through notched joints. These crossbeams constitute the essential building block of FRE fingers. The "front" of the finger is defined as the beam intended to be in contact with the target object although this distinction is arbitrary since the finger is symmetrical. A 3D-printed prototype based on the same dimensions as the commercial product is shown in Fig. 1(b) under load. As it clearly appears, the deformation of the material occurs mostly at the front and back beams as well as the notched joints at both ends of the crossbeams. The crossbeams themselves on the other hand remain relatively undeformed. One can notice in Fig. 1(b) that the bottom crossbeam of a finger is usually clamped by an adapter part (red) to the motion platform on which the finger is attached. This is the recommended attachment method of the finger by the manufacturer. Hence, only eight crossbeams are "active" and actually function in the shape adaptation. The motion platform often takes the form of an industrial parallel gripper. Therein lies also a strength of these soft fingers compared to other soft robotic hands: they take advantage and can use existing hardware with which the industry is very familiar. Thus, the migration from a stiff gripper to a soft robotic alternative is significantly easier.

A brief search online and in the literature for information about grippers using the FRE yields a noticeable amount of images and videos depicting designs that have been implemented in a wide variety of grasping applications such as for fragile agricultural products as shown in Hemming et al. (2014); Eizicovits et al. (2016) but also medical apparatus and cyber-physical systems, see Kölle and Schwarz (2016); Fischinger et al. (2016) for examples. In Kim et al. (2017), the FinGripper of Festo is categorized as a force distributing compliant structure, along with the structures based on similar basic principles: the foot of the StickyBot described in Kim et al. (2008) from Stanford University and the force transferring mechanism from a walking assistive robot by Samsung Electronics as in Lee et al. (2016). On the other hand and quite surprisingly considering its popularity, scientific publications about FRE grippers are rather few and far between. Amongst these published works, a few articles focus on the actual and practical design of a fin gripper, for instance the works of Pfaff et al. (2011); Crooks et al. (2016, 2017); Basson et al. (2017b,a); Hussain et al. (2019); Muddasar (2019) or details on the crossbeam structure in Zapciu, Aurelian et al. (2017); Rosa et al. (2017) but mostly they present FRE fingers as a convenient tool for the validation of other robotic systems, e.g. in Canio et al. (2016); Gandarias et al. (2018); Hemming et al. (2014); Eizicovits et al. (2016). In all these works and to the best of the authors' knowledge, no mathematical modeling has ever been proposed to accurately describe the capability of these fingers to deform and compute the contact forces they are able to generate. This paper aims at filling this gap and accurately predict these fingers' grasp capability with a model combining both shape and force estimation. To this aim, a kinetostatic model is proposed and the best parameters for the equivalent stiffness in its joints are identified. Then, the results of this model with typical metrics used to measure grasp performance are shown to provide some insights into the design of a general fin gripper with an arbitrary geometry. The efficient computational tool thereby made available also opens the door to further optimization of FRE fingers for specific tasks or under particular constraints.

To analyze the kinetostatic behaviour of the general n-crossbeam finger depicted in Fig. 1(a), a pseudo-rigid-body model (PRBM) inspired by the work of Howell (2001) with n degrees of freedom (DOF), n phalanges connected by n deformable crossbeams, and one base crossbeam used for actuation is introduced. Previous PRBM or continuous techniques found in the literature typically focused on the accurate prediction of the position of soft/compliant robotic devices under load based on a modeling of their joints (deformation was restricted to these joints), e.g. by Odhner and Dollar (2012); Venkiteswaran and Su (2016). However, in our case, the back and front of the fingers also significantly deform while having no such clearly identifiable joints. Typical models for soft robotic systems based on the popular piecewise constant curvature (PCC) and Cosserat techniques have also been proposed by Robert J. Webster and Jones (2010); Rucker and III (2011); Renda et al. (2012) but also suffer from some limitations. Cosserat models are arguably more difficult to handle as the PRBM approach proposed here since they are uncommon outside soft robotics and because of their continuum nature, even if discretized versions have been proposed as in Renda et al. (2018). PCC models on the other hand are reported by Renda et al. (2018) to be inaccurate when the soft device they model is subjected to non-negligible external loads as is typical for FRE fingers. Finite element (FE) models constitute a third popular method in soft robotics and have been shown to be able to produce accurate results for predicting the deformation of grasping soft fingers by Coevoet et al. (2019) and conduct shape optimization in Morzadec et al. (2019) but as any model based on FE methods the number of DOF remains large. Workarounds to deal with this issue have been proposed in the literature such as a procedure where the designer progressively refines the parameter space in Morzadec et al. (2019) or algorithmically, as in Goury and Duriez (2018) to reduce the complexity of the model. Finally, our model will also be shown to be able to predict with a fair degree of accuracy both the contact forces along the finger and the overall grasp strength, which none of the previous models dealt with.

The paper is organized as follows: the next Section presents the derivation of a mathematical model of a general FRE finger with n active crossbeams, or n crossbeams for short. Then, a multibody model of a 3-phalanx FRE finger is selected as an example to validate the proposed mathematical model with different sets of contact forces. The validation of the kinetostatic model and the estimation of the stiffness coefficients through finite element analyses (FEA) and experiments with a 3D-printed 8-phalanx FRE finger will subsequently be discussed. Following these experiments, a practical grasp configuration using a pair of FRE fingers will also be validated experimentally using the proposed mathematical model and the estimates of the stiffness coefficients. Finally, appropriate performance metrics will be defined and the impact of the stiffness distribution on the resulting quality of the grasp.

2 Kinetostatic Analysis

The PRBM discussed in this paper is shown in Fig. 2. The living hinges at the vertices of each crossbeam are modeled by revolute joints while additional revolute joints are placed in the front and back beams at the midpoints between each crossbeams, except for the lowest one. This base (lowest) crossbeam of the finger, referred to as the 0th crossbeam, is assumed clamped by an adapter for the installation on a translational actuation device. Hence, the geometric closed-loop containing this proximal crossbeam needs to be considered distinctively from the others. Generally speaking, the ith phalanx on the front beam starts from the middle point between the connections of the $(i-1)^{th}$ and i^{th} crossbeams to the front beam and ends at the middle point between the connections of the ith and $(i+1)^{th}$ crossbeams to the front beam. The "phalanges" of the back beam are defined analogously. In other words, the crossbeams are connected at the middle points of the front and back phalanges as illustrated in Fig 2. At each layer of the model linkage in Fig. 2(b), the links share the same subscript number.

A local frame is attached to the 1st phalanx and centered in O_1^F (see Fig. 1), i.e. the middle point between the connections of the 0th and 1st crossbeams to the front beam, and with its x-direction aligned with the phalanx. One can define a local frame at each front phalanges similarly, see Fig. 2(a). The definition of the linkage geometry is also illustrated in Fig. 1(b). The distal stage of the finger is slightly different as it can be seen from Fig. 1(a) that the distance between the tip of the finger to the distal crossbeam is only around half the distance between other successive crossbeams. If the contact forces are assumed to act at the middle point of the corresponding phalanges, it means that the ith contact force of the ith front phalanx acts at the intersection of the line of the ith crossbeam with that phalanx at the rest configuration, cf. Figs. 1(a) and 2(a). The vector $\boldsymbol{\varepsilon}_i^F$ is then defined from C_i to the nearest joint on the ith crossbeam and has a norm denoted ε_i, as depicted in Fig. 2(b). Similarly, one can define the vector $\boldsymbol{\varepsilon}_i^B$ for the back beam that goes from the middle of the ith back phalanx to the nearest joint on the ith crossbeam. Assuming the

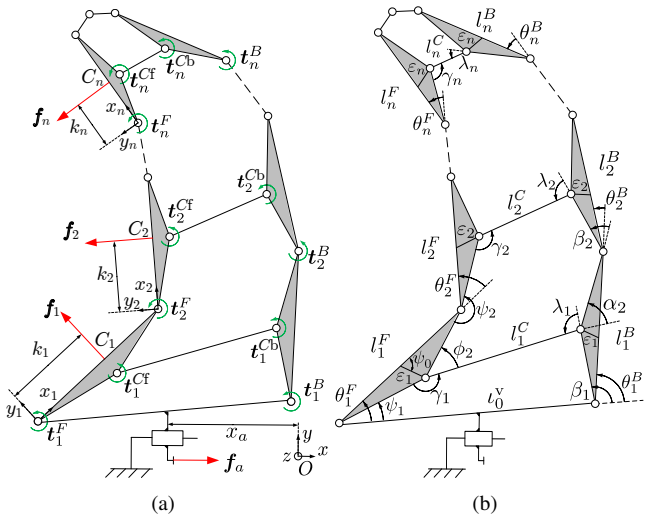


Figure 2. Parameterization of the fin-gripper model: (a) loads; (b) geometry

finger symmetrical, one has: $||\boldsymbol{\varepsilon}_i^F|| = ||\boldsymbol{\varepsilon}_i^B|| = \varepsilon_i$ for all i. The angle between the vector $\boldsymbol{\varepsilon}_i^F$ and the associated phalanx is defined as ψ_0 and is assumed constant and identical for all stages. The relative angle of inclination between the phalanges of the front beam is defined as ψ_i. The inclination of the front beam at rest is defined as ψ_r, which is not equal to ψ_0 unless the finger is installed with the crossbeams horizontal. It should be noted that this model is valid for n phalanges and n crossbeams where n is arbitrary.

The virtual work of our FRE finger model under load can be divided into the work of the contact forces deflecting the front beam, the work of the passive torques deforming the front and back beams at their joints, the work of the passive torques deforming the ends of the crossbeams, and the work of the actuation force on the clamped crossbeam that moves the whole finger. Assuming a translational actuation device, the explicit expression of the total work is:

$$\begin{aligned}\delta W \;=\; & \boldsymbol{f}_a^T\delta\boldsymbol{x}_a + \boldsymbol{f}_F^T\delta\boldsymbol{y}_F + \boldsymbol{t}_F^T\delta\boldsymbol{\theta}_F + \boldsymbol{t}_B^T\delta\boldsymbol{\theta}_B \\ & + \boldsymbol{t}_{C\mathrm{f}}^T\delta\boldsymbol{\theta}_{C\mathrm{f}} + \boldsymbol{t}_{C\mathrm{b}}^T\delta\boldsymbol{\theta}_{C\mathrm{b}},\end{aligned} \tag{1}$$

where $\boldsymbol{f}_a$ and $\delta\boldsymbol{x}_a$ are respectively the force and virtual displacement at the actuation. The subscripts F, B, $C\mathrm{f}$, and $C\mathrm{b}$ stand for, respectively, the front phalanges, the back phalanges, the front-phalanx sides of the crossbeams, and the back-phalanx sides of the crossbeams. The vector $\boldsymbol{f}_F = [f_1\ f_2 \cdots f_n]^T$ represents the contact forces generated by or applied to the finger. These forces are assumed normal to the surface of the phalanges. Each contact force also acts along a vector $\boldsymbol{y}_i$, i.e., $\boldsymbol{f}_i = f_i\boldsymbol{y}_i$, $(i = 1, \ldots, n)$. The vector $\delta\boldsymbol{y}_F$ is defined by $\delta\boldsymbol{y}_F = \left[\delta\boldsymbol{r}_{C_1}^T\ \delta\boldsymbol{r}_{C_2}^T \cdots \delta\boldsymbol{r}_{C_n}^T\right]^T$ where $\boldsymbol{r}_{C_i}$ is the vector from an arbitrarily defined global origin O to the contact point C_i that is itself at a distance k_i from the local origin O_i^F of the associated phalanx. The last four dot products in Eq. (1) represent the virtual work done by the passive elements, i.e. the deformations of the pseudo-rigid joints. Each torque vector $\boldsymbol{t}_x$ $(x = F, B, C\mathrm{f}, C\mathrm{b})$ from the passive elements is defined normal to the motion plane of the finger as shown in Fig. 2(a). The rotations associated with these torques, namely the relative joint angles illustrated

in Fig. 2(b), can be grouped in vectors for convenience. By observing Fig. 2(b) again, the virtual displacements in the relative joint angles can be substituted by:

$$\delta\boldsymbol{\theta}_F = \delta\boldsymbol{\psi}, \quad \delta\boldsymbol{\theta}_B = \delta\boldsymbol{\beta}, \quad \delta\boldsymbol{\theta}_{C\mathrm{f}} = \delta\boldsymbol{\gamma}, \quad \delta\boldsymbol{\theta}_{C\mathrm{b}} = \delta\boldsymbol{\lambda}. \tag{2}$$

Introducing a Jacobian and Transmission matrices defined similarly as in Birglen et al. (2008), the external forces and virtual displacements can be expressed as:

$$\begin{aligned} &\boldsymbol{f} = \begin{bmatrix} f_a \\ \boldsymbol{f}_F \end{bmatrix}, && \begin{bmatrix} \delta x_a \\ \delta\boldsymbol{y}_F \end{bmatrix} = \boldsymbol{J} \begin{bmatrix} \delta x_a \\ \delta\boldsymbol{\theta}_F \end{bmatrix}, \\ &\delta\boldsymbol{\theta}_F = \boldsymbol{T}_F \begin{bmatrix} \delta x_a \\ \delta\boldsymbol{\theta}_F \end{bmatrix}, && \delta\boldsymbol{\theta}_B = \boldsymbol{T}_B \begin{bmatrix} \delta x_a \\ \delta\boldsymbol{\theta}_F \end{bmatrix}, \\ &\delta\boldsymbol{\theta}_{C\mathrm{f}} = \boldsymbol{T}_{C\mathrm{f}} \begin{bmatrix} \delta x_a \\ \delta\boldsymbol{\theta}_F \end{bmatrix}, && \delta\boldsymbol{\theta}_{C\mathrm{b}} = \boldsymbol{T}_{C\mathrm{b}} \begin{bmatrix} \delta x_a \\ \delta\boldsymbol{\theta}_F \end{bmatrix}. \end{aligned} \tag{3}$$

The total virtual work can then be rewritten in terms of the joint angles of solely the front beam and the actuation travel, i.e.:

$$\delta W = (\boldsymbol{f}^T\boldsymbol{J} + \boldsymbol{t}_F^T\boldsymbol{T}_F + \boldsymbol{t}_B^T\boldsymbol{T}_B + \boldsymbol{t}_{C\mathrm{f}}^T\boldsymbol{T}_{C\mathrm{f}} + \boldsymbol{t}_{C\mathrm{b}}^T\boldsymbol{T}_{C\mathrm{b}}) \begin{bmatrix} \delta x_a \\ \delta\boldsymbol{\theta}_F \end{bmatrix}. \tag{4}$$

Now, the expressions of the different matrices in Eq. (4) must be found. Starting with the Jacobian matrix $\boldsymbol{J}$ and with the entry J_{ij} at the ith line and j^{th} column of that matrix, similarly to the method shown in Birglen (2015), one has:

$$J_{11} = 1, \quad J_{i1} = -s_{1\cdots(i-1)}, \quad i = 2, \ldots, n+1, \tag{5}$$

and

$$J_{ij} = \begin{cases} 0, \; i < j \\ k_{i-1}, \; i = j \\ \displaystyle\sum_{\kappa=1}^{i-j} l^F_{j+\kappa-2} c_{(j+\kappa-1)\cdots(i-1)} + k_{i-1}, \; i > j \end{cases} \tag{6}$$

in all other cases. The length l_i^F is the length of the ith phalanx, as shown in Fig. 2(b), and:

$$s_{1\cdots(i-1)} = \sin\left(\sum_{k=1}^{i-1}\theta_k^F\right), \tag{7}$$

$$c_{(j+\kappa-1)\cdots(i-1)} = \cos\left(\sum_{k=j+\kappa-1}^{i-1}\theta_k^F\right). \tag{8}$$

are shorthand notations for sine and cosine of sums of angles.

The expression of the first Transmission matrix, namely $\boldsymbol{T}_F$, is trivially:

$$\boldsymbol{T}_F = [\boldsymbol{0}_{n\times 1} \; \boldsymbol{1}_{n\times n}]. \tag{9}$$

since the corresponding joint angles are selected as the DOF. In the previous equation, $\boldsymbol{0}$ and $\boldsymbol{1}$ denotes the zero and unit matrices respectively.

The other Transmission matrices require a bit more work to express. Except for the proximal and distal geometric closed-loops (layers) of the finger, all other geometric loops of the linkage illustrated in Fig. 2 can be modeled as 6-bar linkages whose parameters are shown in Figs. 3(a) and 3(b). The proximal closed-loop on the other hand is modeled as a 4-bar linkage, as depicted in Figs. 3(c) and 3(d). The unit direction vectors $\boldsymbol{u}_i^X$ and $\boldsymbol{v}_i^X$ along the front ($X = F$) and back ($X = B$) phalanges in these figures are dimensionless, depending only on the joint angles. Finally, the distal closed-loop is modeled as a 5-bar linkage, which allows the mechanism to have the nth and final DOF.

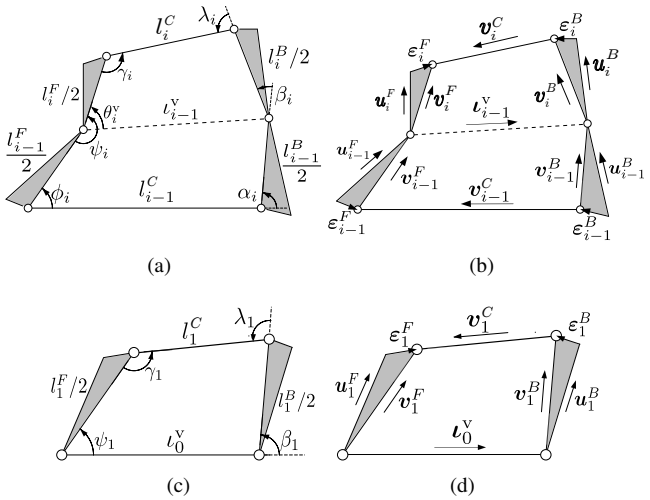


Figure 3. Closed-loop linkages: (a) 6-bar parameters; (b) 6-bar vectors; (c) 4-bar parameters; (d) 4-bar vectors

To derive the Transmission matrices $\boldsymbol{T}_B$, $\boldsymbol{T}_{C\mathrm{f}}$ and $\boldsymbol{T}_{C\mathrm{b}}$ relating $\dot{\boldsymbol{\beta}}$, $\dot{\boldsymbol{\gamma}}$, and $\dot{\boldsymbol{\lambda}}$ to $\dot{\boldsymbol{\psi}}$, the derivatives with respect to time of the closed-loop position equations must be considered for both the 6-bar and 4-bar linkages. This procedure yields:

$$\begin{aligned} &\boldsymbol{A}_i[\dot{\beta}_i \; \dot{\gamma}_i \; \dot{\lambda}_i]^T = \boldsymbol{V}_i\dot{\boldsymbol{q}}_i, \\ &\text{with } \dot{\boldsymbol{q}}_i = \begin{cases} \dot{\psi}_1, & i = 1 \\ [\dot{\phi}_i \; \dot{\alpha}_i \; \dot{\psi}_i]^T, & i > 1 \end{cases} \end{aligned} \tag{10}$$

The detailed expressions of the previous $\boldsymbol{A}_i$ and $\boldsymbol{V}_i$ matrices can be found in Appendix. Once they are are obtained, one has, for $i = 1, \ldots, n$:

$$\dot{\beta}_i = \boldsymbol{x}^T\boldsymbol{W}_i\dot{\boldsymbol{q}}_i, \quad \dot{\gamma}_i = \boldsymbol{y}^T\boldsymbol{W}_i\dot{\boldsymbol{q}}_i, \quad \dot{\lambda}_i = \boldsymbol{z}^T\boldsymbol{W}_i\dot{\boldsymbol{q}}_i \tag{11}$$

where $\boldsymbol{x}$, $\boldsymbol{y}$, and $\boldsymbol{z}$ define the global coordinate frame (i.e. $\boldsymbol{x} = [1\;0\;0]^T$, etc.) and with:

$$\boldsymbol{W}_i = \boldsymbol{A}_i^{-1}\boldsymbol{V}_i. \tag{12}$$

Then, the vector $\dot{\boldsymbol{q}}$ in Eq. (10) needs to be expressed in terms of $\dot{\boldsymbol{\psi}}$, the generalized coordinates. Based on the geometric relationships that can be derived from Fig. 2(b), the derivatives of the angles ϕ_i and α_i can be expressed as

$$\dot{\phi}_i = \dot{\gamma}_{i-1}, \quad \dot{\alpha}_i = \dot{\lambda}_{i-1}, \quad i = 1, \ldots, n. \tag{13}$$

Recursively, the vector $\dot{\boldsymbol{q}}_i$ is finally in the form:

$$\dot{\boldsymbol{q}}_i = \boldsymbol{T}_i^q \begin{bmatrix} \dot{\psi}_1 \\ \dot{\psi}_2 \\ \vdots \\ \dot{\psi}_i \end{bmatrix} \tag{14}$$

with:

$$\boldsymbol{T}_i^q = \begin{cases} 1, & i = 1 \\ \prod_{j=1}^{i-1} \begin{bmatrix} -\boldsymbol{y}^T\boldsymbol{W}_{i-j} & \boldsymbol{0}_{2\times j} \\ -\boldsymbol{x}^T\boldsymbol{W}_{i-j} & \\ \boldsymbol{0}_{j\times\kappa} & \boldsymbol{1}_{j\times j} \end{bmatrix}, & i > 1 \end{cases} \tag{15}$$

and:

$$\kappa = \begin{cases} 3, & j < i-1 \\ 1, & j = i-1 \end{cases} \quad \text{for } i = 1, \ldots, n. \tag{16}$$

Upon substituting Eqs. (14) and (11) into Eq. (2), the Transmission matrices in Eq. (3) can be expressed as:

$$\boldsymbol{T}_j = \begin{bmatrix} 0 & \boldsymbol{\kappa}^T\boldsymbol{W}_1\boldsymbol{T}_1^q & \boldsymbol{0}_{1\times(n-1)} \\ \vdots & & \\ 0 & \boldsymbol{\kappa}^T\boldsymbol{W}_i\boldsymbol{T}_i^q & \boldsymbol{0}_{1\times(n-i)} \\ \vdots & & \\ 0 & \boldsymbol{\kappa}^T\boldsymbol{W}_n\boldsymbol{T}_n^q & \end{bmatrix}, \boldsymbol{\kappa} = \begin{cases} \boldsymbol{x}, & j = B \\ \boldsymbol{y}, & j = C\mathrm{f} \\ \boldsymbol{z}, & j = C\mathrm{b} \end{cases} \tag{17}$$

Finally, coming back to the expression of the virtual work defined in Eq. (1) and equating it to zero, one can obtain the expression of the contact forces at equilibrium as:

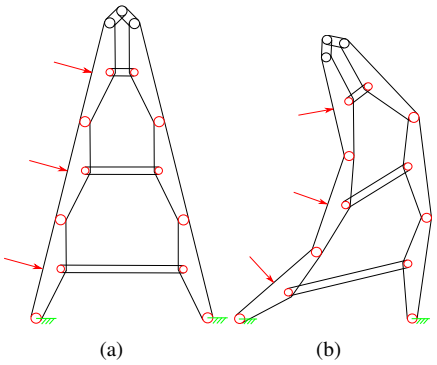
$$\boldsymbol{f} = -\boldsymbol{J}^{-T}(\boldsymbol{T}_B^T\boldsymbol{t}_B + \boldsymbol{T}_F^T\boldsymbol{t}_F + \boldsymbol{T}_{C\mathrm{f}}^T\boldsymbol{t}_{C\mathrm{f}} + \boldsymbol{T}_{C\mathrm{b}}^T\boldsymbol{t}_{C\mathrm{b}}). \tag{18}$$

A geometric analysis can then be conducted to calculate the back-beam angles producing the torque $\boldsymbol{t}_B$ as well as the crossbeam angles producing $\boldsymbol{t}_{C\mathrm{f}}$ and $\boldsymbol{t}_{C\mathrm{b}}$ from the set of inputs, namely the front-beam angles. As can be seen from Fig. 2, when the links from the ith layer are fixed, the $(i+1)^{th}$ layer forms a simple four-bar linkage and thus, the back-beam joint angle of the $(i+1)^{th}$ layer can be expressed by means of simple geometric relationships, see Appendix for detailed expressions.

3 Multibody Simulations and Validation

A model in MSC Adams, a commercially available dynamic simulation package, was then constructed to simulate the motion of a 3-phalanx FRE finger based on the schematic diagram proposed in Fig. 2. This model is shown in Fig. 4. The results from the software can then be compared with the calculations presented in the previous section. To this aim, three contact forces were applied at the mid-points of the front phalanges, while 12 torsional springs, indicated by red circles* in Fig. 4, were added at the joints of the front and back phalanges as well as the deformable crossbeams.

One should notice that instead of applying an actuation force at the base of the finger, concentrated forces perpendicular to the phalanges were applied to obtain the corresponding reaction forces and deformation of the finger. This simple modification allows for a much faster simulation time since both the computations pertaining to the translation axis and the contact model are removed. Indeed, setting the right parameters for a contact between two solids in Adams requires many adjustments (e.g. penetration depth, energy restitution coefficients, etc.) that are needed for proper dynamic modeling but irrelevant in quasi-statics. Directly applying contact forces allows to bypass these tedious settings and speed up the simulation.



(a) (b)

Figure 4. MSC Adams multibody dynamic model: (a) at rest; (b) deformed

The explicit expressions of the Jacobian and Transmission matrices of this 3-DOF FRE finger as obtained from our model are:

$$\boldsymbol{J} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -s_1 & k_1 & 0 & 0 \\ -s_{12} & l_1^F c_2 + k_2 & k_2 & 0 \\ -s_{123} & l_1^F c_{23} + l_2^F c_3 + k_3 & l_2^F c_3 + k_3 & k_3 \end{bmatrix},$$

$$\boldsymbol{T}_F = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{T}_B = \begin{bmatrix} 0 & \boldsymbol{z}^T\boldsymbol{W}_1 & 0 & 0 \\ 0 & \boldsymbol{z}^T\boldsymbol{W}_2\boldsymbol{T}_2^q & 0 & \\ 0 & & \boldsymbol{z}^T\boldsymbol{W}_3\boldsymbol{T}_3^q & \end{bmatrix},$$

$$\boldsymbol{T}_{C\mathrm{f}} = \begin{bmatrix} 0 & \boldsymbol{y}^T\boldsymbol{W}_1 & 0 & 0 \\ 0 & \boldsymbol{y}^T\boldsymbol{W}_2\boldsymbol{T}_2^q & 0 & \\ 0 & & \boldsymbol{y}^T\boldsymbol{W}_3\boldsymbol{T}_3^q & \end{bmatrix}, \boldsymbol{T}_{C\mathrm{b}} = \begin{bmatrix} 0 & \boldsymbol{x}^T\boldsymbol{W}_1 & 0 & 0 \\ 0 & \boldsymbol{x}^T\boldsymbol{W}_2\boldsymbol{T}_2^q & 0 & \\ 0 & & \boldsymbol{x}^T\boldsymbol{W}_3\boldsymbol{T}_3^q & \end{bmatrix} \tag{19}$$

with

$$\begin{aligned} \boldsymbol{T}_2^q &= \begin{bmatrix} -\boldsymbol{y}^T\boldsymbol{W}_1 & 0 \\ -\boldsymbol{x}^T\boldsymbol{W}_1 & 0 \\ 0 & 1 \end{bmatrix}, \\ \boldsymbol{T}_3^q &= \begin{bmatrix} -\boldsymbol{y}^T\boldsymbol{W}_2 & 0 \\ -\boldsymbol{x}^T\boldsymbol{W}_2 & 0 \\ \boldsymbol{0}_{1\times 2} & 1 \end{bmatrix} \begin{bmatrix} -\boldsymbol{y}^T\boldsymbol{W}_1 & \boldsymbol{0}_{2\times 2} \\ -\boldsymbol{x}^T\boldsymbol{W}_1 & \\ \boldsymbol{0}_{2\times 1} & \boldsymbol{1}_2 \end{bmatrix} \end{aligned} \tag{20}$$

and $\boldsymbol{W}_1 = \boldsymbol{A}_1^{-1}\boldsymbol{V}_1$, $\boldsymbol{W}_2 = \boldsymbol{A}_2^{-1}\boldsymbol{V}_2$, $\boldsymbol{W}_3 = \boldsymbol{A}_3^{-1}\boldsymbol{V}_3$. The detailed expressions of $\boldsymbol{W}_i$, $\boldsymbol{A}_i$ and $\boldsymbol{V}_i$ for $i = 1, 2, 3$, can be found in Appendix.

The specific parameters of the model depicted in Fig. 4 are listed in Table 1, where $K_i^x = t_i^x/\Delta\theta_i^x$, $x = F, B, C\mathrm{f}, C\mathrm{b}$ are the stiffnesses at each joint linearly relating the compliance torques to the deformation angles. These stiffness coefficients were not defined in the previous Section per se to allow for generalization of these torques. For instance, if nonlinear torques with respect to joint angles are required, such as model could be used. The geometric parameters were set to keep the overall proportion of the FRE finger depicted in Fig. 1(a). The actual values of the stiffness coefficients set in Adams were chosen to obtain a realistic

*The standard symbol for torsional springs in Adams is replaced here by red circles to increase the readability of the figure with a dozen torsional springs.

Table 1. Stiffness coefficients (Nmm/°) and Structural parameters

Stiffness	Values	Geometry	Values
$K_i^F,\ i=1,2,3$	50	$l_i^F,\ i=1,2,3$	412.3 mm
$K_i^B,\ i=1,2,3$	50	$l_i^B,\ i=1,2,3$	412.3 mm
$K_i^{C\mathrm{f}},\ i=1,2$	20	l_0^C	700 mm
$K_i^{C\mathrm{b}},\ i=1,2$	20	l_1^C	500 mm
$K_3^{C\mathrm{f}}$	100	l_2^C	300 mm
$K_3^{C\mathrm{b}}$	100	l_3^C	100 mm
		$\varepsilon_i,\ i=1,2,3$	50 mm
		$\psi_r=\psi_0$	76°

deformation by trial and error. The forces applied in this example as well as the comparison between our model and the numerical results from Adams are listed in Table 2 for five sets of arbitrary forces. The differences between both are listed as "errors" in this table. As it can be seen, these errors are negligible and therefore, the proposed mathematical model is deemed valid for a defined set of kinetostatic parameters. The following section then investigates the applicability of the proposed model to its more complex soft counterparts and more specifically, discusses a more realistic determination of the stiffness coefficients to yield accurate results compared to experiments.

4 Finite Element Analysis and Stiffness Estimation

If the geometric parameters of our kinetostatic model for FRE fingers can be readily obtained by observation of these fingers, the equivalent stiffnesses of this lumped compliance model are more difficult to estimate. While there exist many analytic and/or semi-empirical formulae for establishing the equivalent stiffness of living and notch hinges with different geometries, see the books of Howell (2001) or Lobontiu (2002) for a list, models for beams with distributed compliance undergoing large deformations and, crucially, in contact with external objects are uncommon, see the work of Steutel et al. (Montreal, Quebec, Canada, 2010) for one of these rare examples. Additionally, most of these formulae are only valid for small deformations which is rarely the case here. Therefore, and since there appeared to be no clearly established technique to use, the estimation of these stiffness coefficients is presented here based on finite element analyses (FEAs) and supported by experimental measurements. The results of these FEA must match the practical experiments, and thus, many different grasps with objects of different sizes were achieved with the 3D-printed flexible finger shown in Fig. 1(b).

It should be emphasized that the kinetostatic model presented in the previous section of this paper generates two sets of crucial information from the n known joint angles of the front phalanges. The first set includes the $3n$ unknown joint angles of the back phalanges and crossbeams, which constitute the predicted shape of the finger. In other words, one can obtain the configuration of the FRE finger simply from the measurement of the joint angles of the front phalanges, which is also a prerequisite for the estimation of the joint torques of the back phalanges and crossbeams. The other set of results comprises the n unknown contact forces and the actuation force, on the condition that the $4n$ stiffness coefficients are known, or more generally, the $4n$ input torques at each joint of the finger are available whether from measurements if sensors can be embedded or from a compliance model as discussed here. Because the direct measurement of the contact forces is rather challenging since it requires adding external sensors, the estimation of the contact forces based on geometric parameters and material properties becomes a valuable tool to the designer and user of FRE fingers.

To validate the shape estimation provided by our model, experimental measurements of the location and size of the target object to grasp as well as the displacement of the fingers are input into the FE model to provide a comparison basis between the FEA and these experiments. The result of these comparisons is that although the overall contact forces along the x- and y-axes cannot be immediately determined because of the unknown Young modulus of the 3D-printing material, the ratio between the projection of the contact forces along these two axes remains identical for each configuration. The specifications of the 3D-printed material was not made available by the manufacturer. Thus, the Young modulus of the 3D-printed material was estimated based on the concordance between the experiments and FEA results. Then, the joint angles of the front phalanges were obtained from the FEA and inputed into the kinetostatic model to compute all the other joint angles. The validation of the contact forces on the other hand requires an estimation of the stiffness coefficients. Assuming a constant scaling factor for the stiffness coefficients with respect to the Young modulus, the second moment of inertia, and the length of the pseudo-rigid joints as typical from the literature was first tried out but did not provide usable results since the FEA was unable to converge to a solution for a variety of configurations. Thus, a new set of stiffness coefficients needs to be found and the approach proposed here is to optimize the latter to fit a wide range of simulation data obtained from FEA. More precisely, a data set of 80 configurations each yielding 9 forces (contacts + actuation) was used to estimate the 16 unknown stiffness coefficients. It should be emphasized that the optimal solution found using this methodology would not provide an accurate estimation of the generated forces unless the proposed kinetostatic approach is valid as will indeed be shown.

In Fig. 6(a), the front beam is discretized into eight phalanges in the FEA software and the proximal one is connected to the base crossbeam clamped by the adapter. The dimensions of the finger, presented in Fig. 1(a), are not shown again in Fig. 6(a) to improve legibility. The contact area of this configuration at equilibrium is shown in Fig. 6(b). The color codes produced by the FEA software divide the status of the simulated contact into five types. The pink, orange, and red areas indicate that the finger is in physical contact with the object, the yellow area means that the finger is very close to the object but not in contact while finally, the blue area points to a relatively greater distance from the object than the yellow area. The pink and red areas highlight that

Table 2. Force estimation (N)

Force	Set #1	Set #2	Set #3	Set #4	Set #5	μ_{row}
f_a	-42.04 (-0.26)	-27.96 (-0.05)	-37.46 (-0.22)	-45.47 (-0.38)	-27.84 (-0.14)	0.54%
f_1	-25 (-0.10)	-15 $(+0.05)$	0 $(+0.05)$	-20 (-0.52)	0 $(+0.23)$	1.11%
f_2	-18 (-0.10)	-10 (-0.12)	-25 (-0.03)	0 $(+0.26)$	-30 (-0.19)	0.63%
f_3	-9 (-0.02)	-7 (-0.08)	-15 (-0.21)	-30 (-0.18)	0 (-0.07)	0.85%
μ_{sim}	0.44%	0.72%	0.53%	1.34%	0.56%	0.78%

Absolute differences with MSC Adams are listed between parentheses.
The percentage errors for the 0 N forces are not considered in the mean value as they lead to division by zero.
μ_{row}: the mean value of the percent errors per row;
μ_{sim}: the mean value of the percent errors per simulation (set).

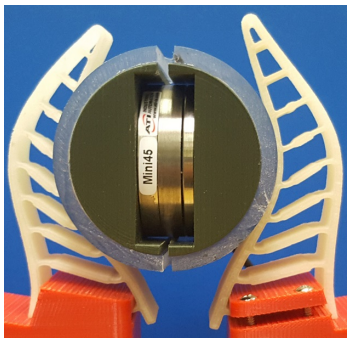


Figure 5. Example of a practical grasp configuration using a pair of FRE fingers and a cylindrical object made of two half shells connected by a force sensor

a contact there is respectively over-constrained and sticking which are ideal in grasping situations but even an orange contact area indicating possible sliding can be sufficient to grasp an object if the finger is enveloping that object. Indeed, the sliding is caused by the deformation of the finger around the object and it does not mean that the latter is moving out of the grasp. The contact area shown in Fig. 6(b) is continuous around the mid-lower part of the front beam. Figure 6(c) illustrates the deformation of the finger and the overall contact force direction and magnitude as applied to the object. The red part at the bottom and the green half cylinder on the left shown in the latter figure represent the adapter for the gripper and a half shell of the cylindrical object both in contact with the FRE finger as showin in Fig. 5. The estimated compression force along the x-axis and the ejection force along the y-axis thereby obtained are 27.3 N and 9.04 N respectively.

Figure 6(d) shows the shape of the finger predicted by our kinematic model in blue solid lines. The inputs of the model are the eight joint angles of the discretized phalanges of the front beam, while the outputs are the 24 joint angles of the back phalanges and of the crossbeams. A minor discrepancy between the solid blue links from these calculations and the dotted purple links representing the shape computed by the FEA can be seen at the distal tip of the back beam. Its magnitude is however quite small. It is therefore reasonable to assume that the kinematics of the mathematical model accurately estimates the finger configuration.

To estimate the actual eight contact forces from the

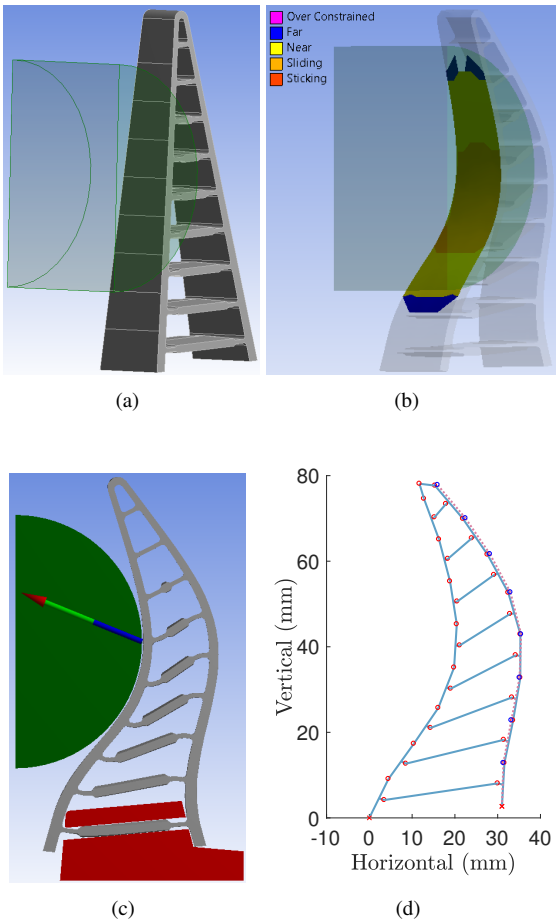


Figure 6. FE contact analysis: (a) discretization of the front beam; (b) contact area under grasp; (c) grasp configuration and composite-force direction; (d) comparison between the FEA and mathematical models

eight input joint angles using the mathematical model, all 32 stiffness coefficients are needed. Since the design of the finger is symmetrical, it is assumed at each layer that the front joint stiffness is equal to the back joint stiffness and that the two ends of the crossbeams similarly share an identical stiffness coefficient. Based on this symmetry, the unknown stiffness values can be reduced to 16. To obtain these 16 coefficients, a numerical approach is introduced in this paper to find them by optimization. This optimization treats the stiffness coefficients as unknowns while the target function to minimize is the sum of the magnitudes of the differences between the contact/actuation forces obtained from our

model and the FEA results. Therefore, the minimization is trying to find the best values of the stiffness coefficients so that the kinetostatic and FEA results are the closest possible. These differences used in the target function are computed for a wide set of 80 configurations of the finger, representing the grasp of objects of different sizes and positions. In the model proposed here, one stiffness coefficient has to be found for each joint of the linkage. Thus, since the number of parameters to estimate is large and initial tries showed that many local minima exist, a genetic algorithm was used to find these best estimates. The standard genetic algorithm in Matlab was utilized to this aim.

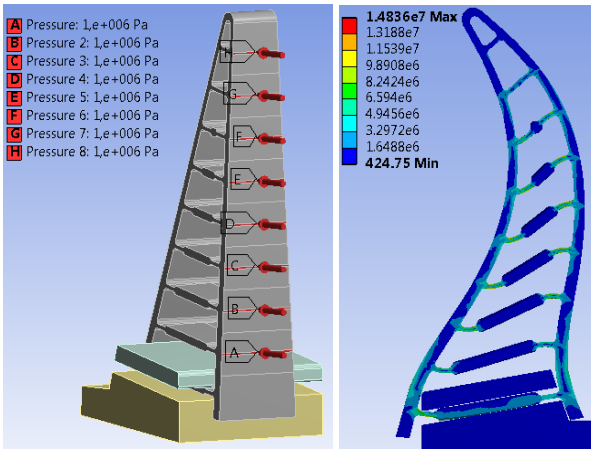


Figure 7. FE pressure analysis: (a) applied pressure at mid-phalanges; (b) resulting stress and deformation

An example of one of the test configurations is shown in Fig. 7. Contact forces are applied on the front phalanges to model the distributed pressures on the continuous contact area. These pressures range between 10^{-6} to 10^{6} Pa and the contact forces f_1 to f_8 are adjusted manually to match the FEA results of the continuous model such as the one in Fig. 6(b). The last force, f_a from the actuation, is obtained from the FEA. The resulting 80 sets of concentrated contact and actuation forces inputed to the FEA software are plotted in Fig. 8. Each line in this plot corresponds to a particular configuration under scrutiny. As can be seen, while the actuation force ranges from a little below 20 N to almost 40 N, the actual contact forces at the phalanges have a significant smaller range of magnitudes, typically less than 10 N. This is typical from these FRE fingers: they provide a rather small contact to actuation force ratio, i.e. a low mechanical advantage. Please note that in Fig. 8 a magnitude of 0 N indicates that no contact occurs at this point.

The optimal stiffness values found using these sets of data by the genetic algorithm are then plotted in Fig. 9. For each layer (index number on the x-axis), two sets of stiffness coefficients are obtained: one for the crossbeam and the other for the front/back beams. Surprisingly, the best stiffness coefficient appear to be roughly proportional to the width and the square of the thickness at the joint. It can also be clearly seen that a constant stiffness model such as the ones classically proposed in the literature is far from optimal (it would correspond to horizontal lines in the Fig. 9). The exact stiffness values resulting from the optimization are listed in Table 3, followed by the plots of the estimation errors of the forces in Fig. 10. In this figure, the estimation errors of all the contact forces and the actuation force are shown to be within 3 N. The percentage errors of the overall compression force are shown in Fig. 11. Statistical indicators on the compression force estimations from both the model and FEA are listed in Table 4.

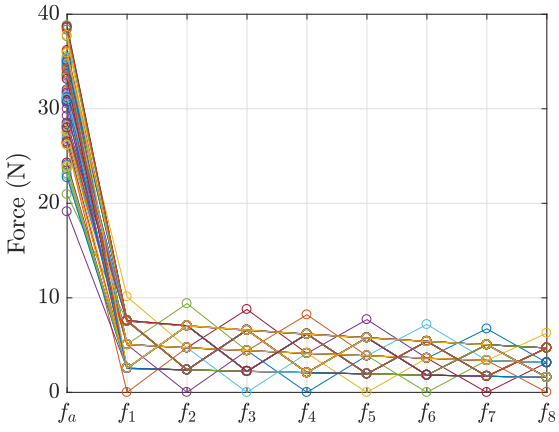


Figure 8. Input forces of the FEA (80 sets) on the eight phalanges

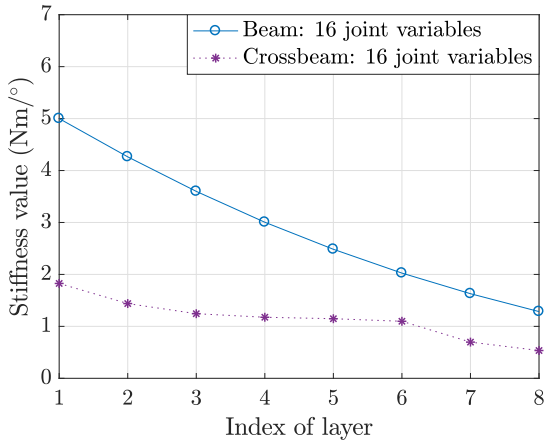


Figure 9. Estimation of the 16 stiffness coefficients for the 32 joints

Table 3. Stiffness values (Nmm/°)

Layer	1	2	3	4	5	6	7	8
Beam	5.01	4.26	3.58	2.97	2.45	1.99	1.59	1.24
Crossbeam	1.83	1.44	1.24	1.18	1.15	1.10	0.70	0.54

Table 4. Statistics of the compression force errors

Statistics	Median	Max	Min	Std
Absolute (N)	0.71	2.08	0.01	0.92
Percentage (%)	2.23	6.55	0.03	3.07

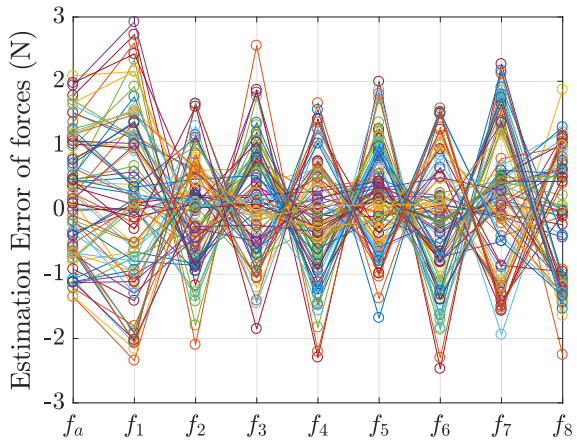


Figure 10. Error of the forces between the best estimate and FEA

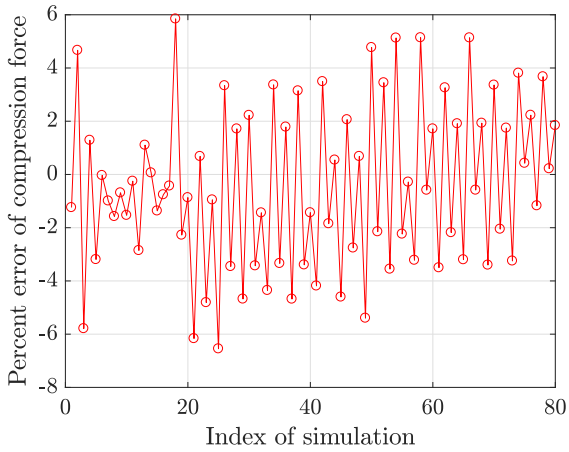


Figure 11. Percentage errors of the compression forces between the best estimate and FEA

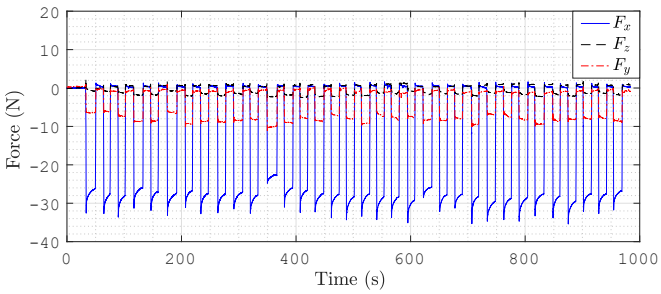


Figure 12. Experimental grasp forces under translational open-and-close motions of the FRE fingers (30 repetitions)

Table 5. Experimental grasp strength comparison (N)

Statistics	Median	Mean	Min	Max	Std	FEA
x-axis	27.6	27.2	22.5	29.7	1.40	27.3
y-axis	8.43	8.24	6.00	10.2	1.00	9.04

Table 6. Contact forces in Analytical vs FE model (N)

Force	Analytical	Concentrated FE
f_a	27.3	27.6
f_1	6.65×10^{-5}	1.37×10^{-3}
f_2	1.63×10^{-5}	4.22×10^{-3}
f_3	6.92	6.26
f_4	17.7	15.7
f_5	7.76	6.94
f_6	0.75	0.681
f_7	3.7×10^{-4}	2.75×10^{-4}
f_8	1.07×10^{-3}	3.40×10^{-4}

5 Validation and Discussion

Now that the results of the proposed model were shown to be close to that of the FEA and as a final step to ensure the accuracy of our model, additional practical experiments were conducted to measure the actual grasp strength (overall compression force) and compare to the one predicted using the proposed technique. According to the test benchmarks proposed by the National Institute of Standards and Technology (NIST) in Falco et al. (2015), grasp strength is an important metric of the performance of robotic hands. The hand we used in our experiment is the same as the one illustrated in the previous section, namely two 3D-printed FRE fingers mounted on an industrial pneumatic gripper (Schunk KGG 100-80) as shown in Fig. 5. Repetitive open-and-close operations at the nominal 6-bar pressure were produced and the resulting grasp strength was measured with a 6-axis ATI Mini45 F/T sensor. The measured grasp strength for the object illustrated in Fig. 5 is plotted in Fig. 12. Table 5 lists the statistics of the measurements as well as the overall grasp forces in the x- and y-axes from FEA. Both are shown to be in good accordance.

Based on the same technique used in the previous Sections of the paper, the FEA contact pressures on the surfaces of the phalanges were transformed to concentrated forces in the same manner as the model in Fig. 7(a). The contact forces from the analytical model and their concentrated counterparts are then compared in Table 6. As is shown, these forces differ slightly while the overall compression forces are very close. It is conjectured that this discrepancy in the contact forces mainly stems from the transformation from the continuously varying pressure distribution within the contact area, as in Fig. 6(b), to the discretized forces. Indeed, the equivalent force locations cannot be decided precisely in a continuous contact situation, which may be a source of errors in the calculation of the Jacobian matrix of the mathematical model. In addition to the similarity of the forces, the joint angles were also verified to be very close in all simulations. The largest angle error is only 1.66° in all 32 joints for all 80 configurations. This makes the FE model with the concentrated forces an acceptable representation of its continuous contact counterpart and by extension, it validates the mathematical model proposed in in this work, which was shown to be close to that representation.

Table 7 also compares the ranges of the joint angles of the

Table 7. Angle range (°)

Joint angle	80 training sets	Test config.
θ_1^F	68.1–79.7	68.7
θ_2^F	-7.09—2.62	-9.56
θ_3^F	-0.25–3.92	1.39
θ_4^F	3.07–9.16	13.0
θ_5^F	4.72–11.6	17.9
θ_6^F	5.35–12.5	12.5
θ_7^F	4.71–11.3	5.76
θ_8^F	4.39–11.0	5.59

80 sets of simulations with an additional test configuration in FEA used for validation. This test configuration does not belong to the ones used in the genetic algorithm optimization. One can observe that three out of the eight joint angles of this new configuration actually go beyond the range of the 80 sets of simulations. The results of the predictions for the contact forces from our model with the parameters of the test configuration are plotted in Fig. 13. The overall actuation and contact forces are clearly quite close. The contact forces close to zero around the finger tip corresponds to no contact or a very small contact area. The percentage error of the compression force is 12.0%, which is noticeably larger than the 6.55% maximum percent error of the 80 sets of simulations in Fig. 11 but not excessively so.

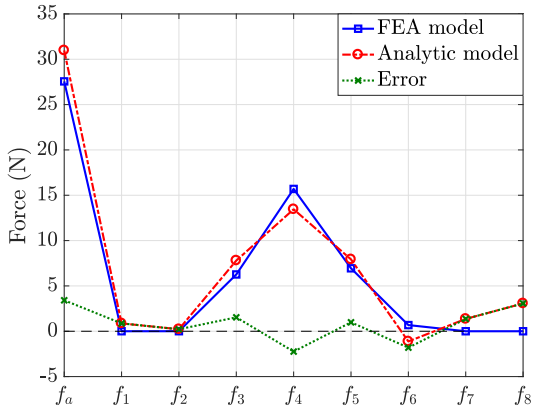


Figure 13. Force estimation and comparison for the reference configuration

The errors from our model are conjectured to be coming from a combination of several sources: firstly, the curvature of a phalanx brings in estimation error of the stiffness, which will result in inaccuracy in the joint torques. Secondly, the deformation of the crossbeams tends to become more "S"-shaped when undergoing large deformations of the finger. These large deformations make the equivalent lengths of the crossbeams shorter than the nominal values. Thirdly, the lengths ε_i regarded rigid at the ends of the crossbeams close to the front/back beam significantly affect the estimation of the joint angles of the crossbeams but are difficult to measure precisely. Last but not least, the front side of the base crossbeam also bends to a certain extend and does not stay entirely undeformed under the clamp of the adapter as in the model, which can be observed in Fig. 7(b). This additional deformation corresponds to a torque that was neglected in the calculation of the virtual work.

To summarize, although the new test case was not used for the estimation of the stiffness coefficients and went beyond the range of the FEA, it was shown that our model still provided a reasonably accurate estimation of the deformed configuration as well as the overall contact forces along x- and y-axes. Moreover, the calculation based on the mathematical model also provides a good estimate of the contact forces.

A series of five artifacts from 25.4 mm to 76.2 mm in diameter as well as a cubic artifact of 45 mm side length were also used in practical grasping experiments and positioned at different heights relatively to the base of the fingers. The cylindrical artifacts typically generate relatively large contact areas in power grasps while the cubic artifact generally exhibits (one or two) narrow contact areas even when this object is located close to the gripper's palm. Noticeably, even with this limited surface of contact, large deformations of the finger can be observed. The combinations of these experimental artifacts held at various heights relatively to the fingers' base yield a wide range of contact heights, depths, area widths and distances which basically stand for a variety of grasp configurations for arbitrary target objects. The experimental configurations were simulated and estimated with both the FEA and mathematical models as shown in Figs. 14 to 23. The resulting grasp forces are compared in Table 8 for the different locations and sizes of the artifacts. In the lower right plot of each figure showing the equivalent links and joints, the blue solid lines from the mathematical model are again shown to match very well the dotted red lines obtained from the FEA displacements.

Table 8 shows that the resulting compression forces obtained from the FE and mathematical (kinetostatic) models are close to the experimental results with some discrepancies of limited magnitudes. Typically, the kinetostatic model tends to overestimate the grasp forces especially for grasps on the low end of the compression force spectrum, e.g. for smaller objects and/or in precision grasps but these deviations remain acceptable for design comparison purposes.

It should be noted that a few lengths of the phalanges of the back beam might need to be adjusted when the finger material "ripples" on the back beam, such as the top phalanges in the configurations D2-5H115, D3H104 and D3H120. The decrease of the equivalent length of the back phalanges, though not dramatic from the illustrated observations, could affect the calculations of the joint angles and hence, the grasp forces which could explain some of the observed differences with the kinetostatic model. Indeed, the latter model disregards internal strains and potential buckling resulting from these strains. A more refined model could include the elongation/compression of the beams themselves, for instance with a virtual prismatic joint whose virtual work would then need to be added to Eq. (1) and

the derivative of the associated translational coordinates expressed as a matrix product with the finger DOF similarly to Eq. (3). However, this would substantially increase the kinematic equations since each stage of the finger would then need to be modelled by an 8-bar linkage. There is no theoretical limit to the number of degrees of freedom of the model linkage but as illustrated by the previous results, the one proposed here although not perfect seems a reasonable compromise between accuracy and complexity.

One should also pay attention to the adapter connection with the finger. It is clear from Fig. 25(b) that stress concentration occurs at the proximal end of the front phalanx in contact with the adapter where the bottom joint (O_1^F) of the kinetostatic model was defined. Moreover, the hinge of the bottom crossbeam closest to O_1^F and clamped by the adapter also shows non-negligible stress and deformation. This phenomenon can also be found from the FEAs in the upper right plots of Figs. 14 to 23, especially during large deformations of the fingers. This is an interesting finding because the deformation of the aforementioned hinge is ”hidden” by the adapter during the experiments and usually not clear to an observer. This deformation can be negligible or not depending on the type of constraint exerted by the adapter on the finger it clamps. The internal torque created there also could be included in a more refined mathematical modeling similarly to the elongation or compression of the beams. However, the shape and clamping mechanism of the adapter then have to be taken into account and since this part is typically specifically designed to match the gripper at hand, any general lesson might be difficult to learn. The current kinetostatic model presented in this paper is more general in that sense.

From the experiments and simulation conducted in this section, it seems beneficial to devise either a short adapter to allow larger bending of the finger though the deformation of the aforementioned hinge should be taken into account or a negative shape adapter to completely constrain the last layer of the finger (both the crossbeam and the phalanges) which in turn should also create stronger grasp forces.

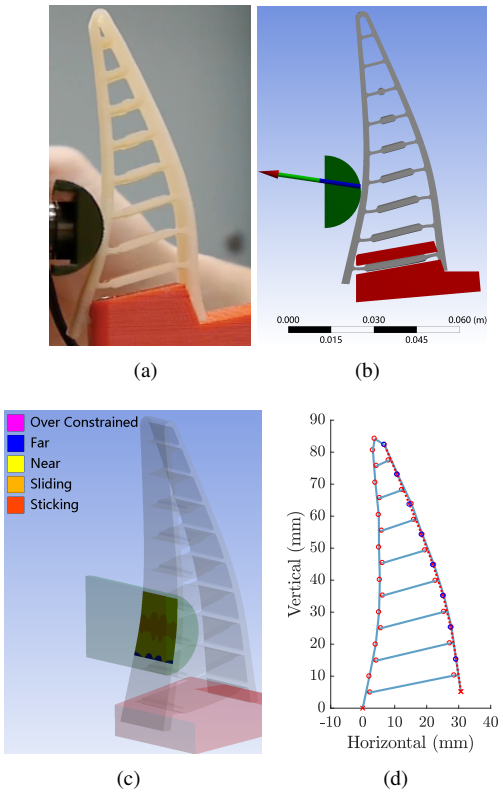


Figure 14. Configuration D1H73: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

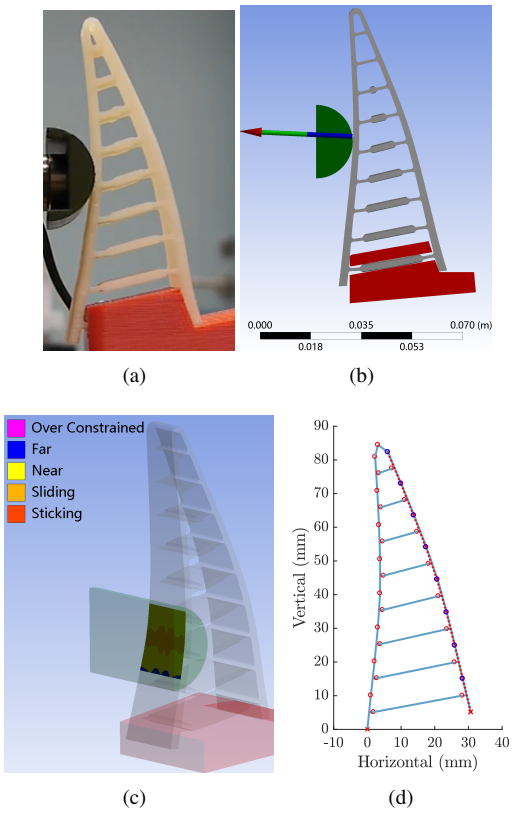


Figure 15. Configuration D1H86: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

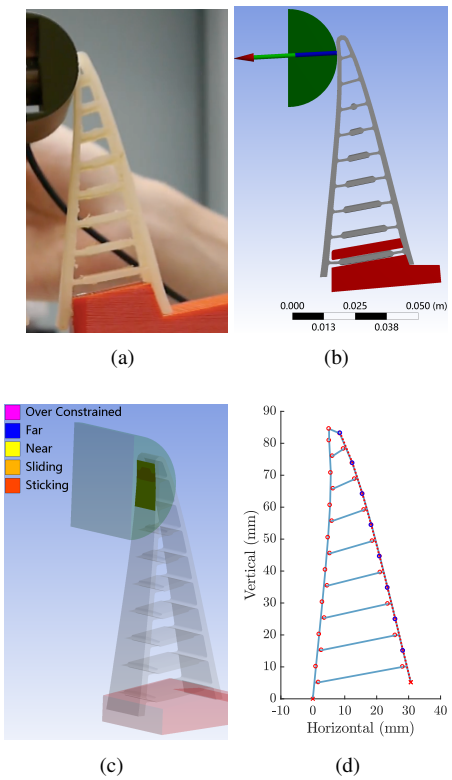


Figure 16. Configuration D1-5H122: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

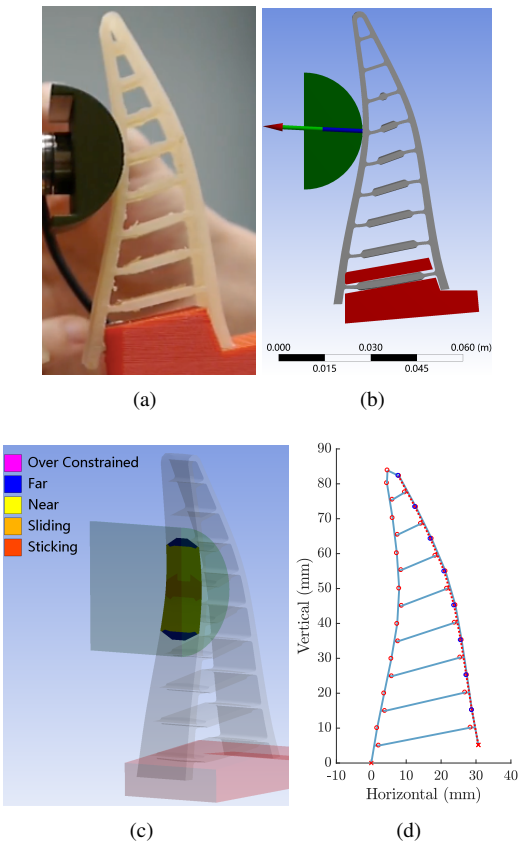


Figure 17. Configuration D1-5H105: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

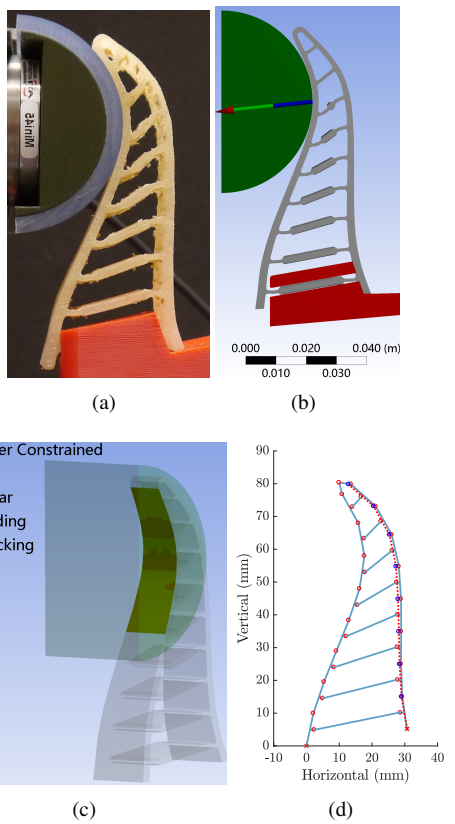


Figure 18. Configuration D2H110: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

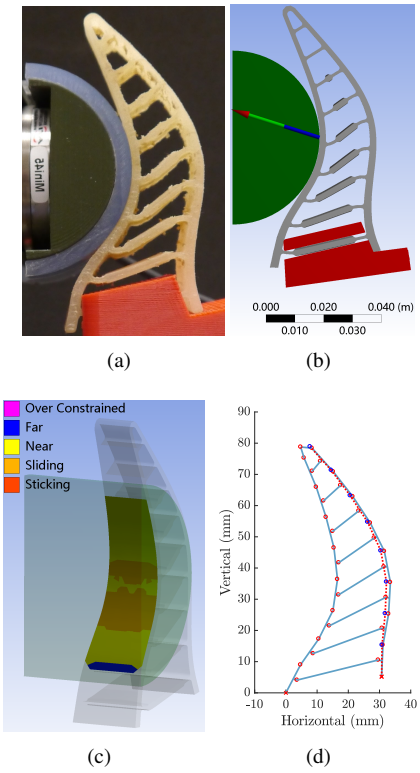


Figure 19. Configuration D2H90: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

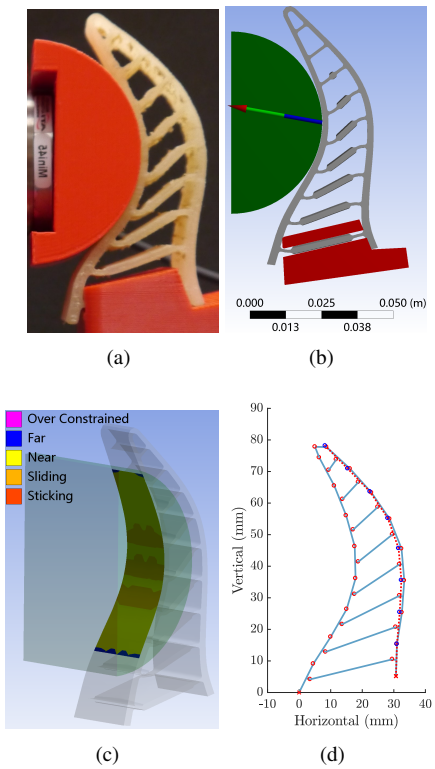


Figure 20. Configuration D2-5H95: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

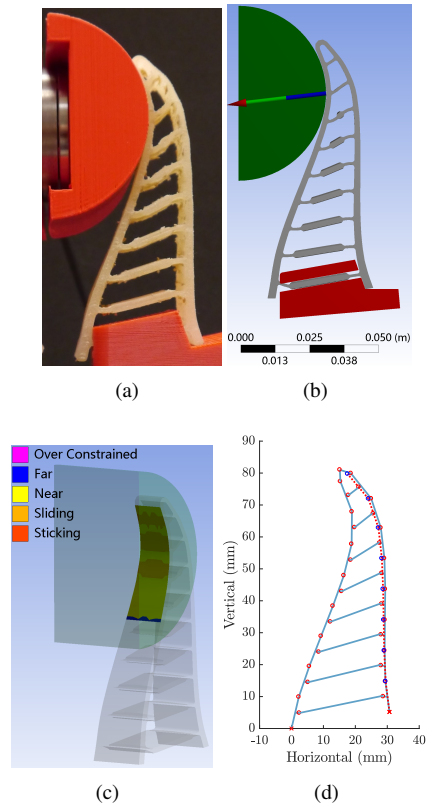


Figure 21. Configuration D2-5H115: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

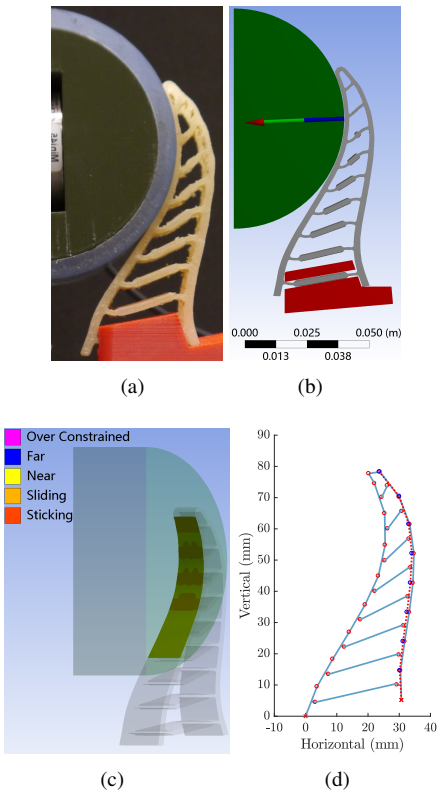


Figure 22. Configuration D3H104: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

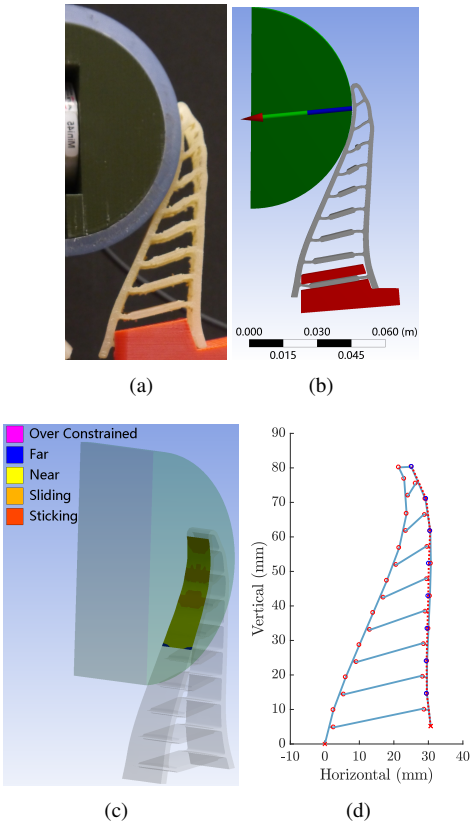


Figure 23. Configuration D3H120: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

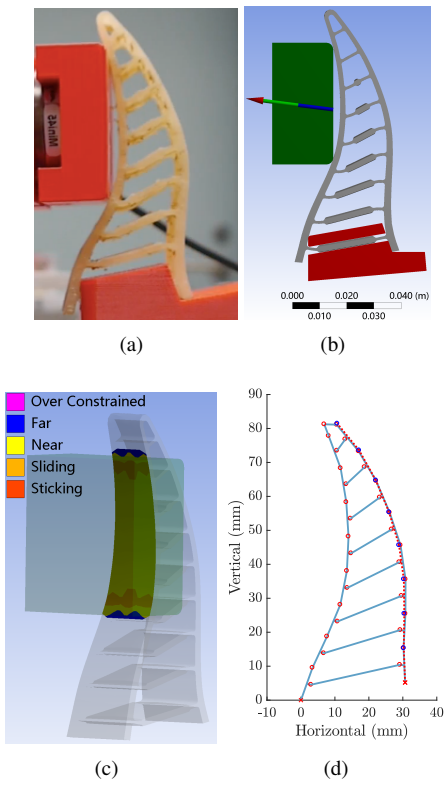


Figure 24. Configuration L45H98: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

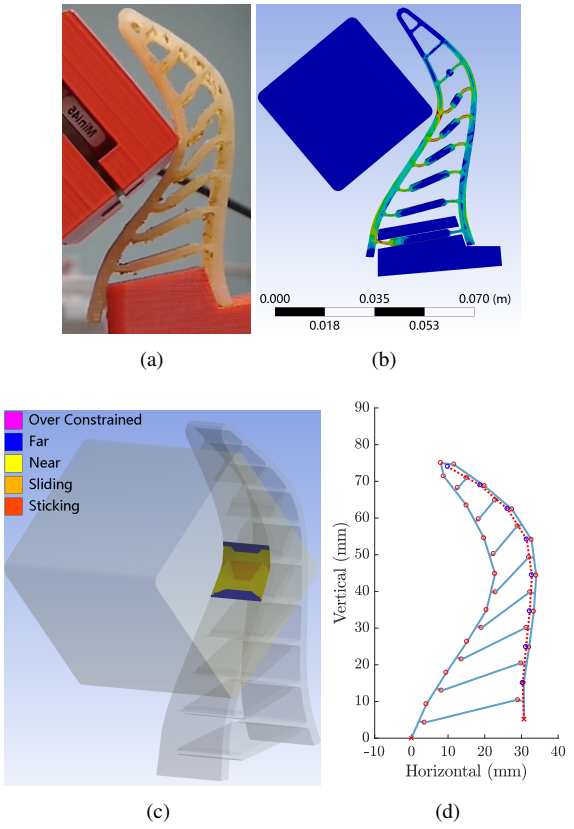


Figure 25. Configuration L45H101: (a) experimental configuration; (b) FEA deformation; (c) contact status; (d) comparison between the FEA and mathematical models

Table 8. Comparison of the compression force

H (mm)	Dims. (mm)	Expt.	FEA	Kinet.
73	25.4	-7.82	-7.87	-9.57
86	25.4	-3.31	-3.54	-4.58
122	38.1	-7.89	-7.83	-9.57
105	38.1	-7.90	-7.88	-8.34
110	50.8	-13.92	-13.80	-13.40
90	50.8	-25.48	-25.44	-29.80
95	63.5	-25.04	-25.21	-29.55
115	63.5	-15.52	-15.62	-16.11
104	76.2	-47.61	-47.17	-48.87
120	76.2	-27.84	-27.74	-28.74
98	45	-23.21	-23.54	-26.36
94	45	-18.10	-18.95	-20.68

Note: H and Dims. stands for the height and the diameter/side-length of the target object, respectively.

6 Grasp Performance Evaluation Example

With the mathematical model established and validated in the previous sections, one can use it with confidence to simulate and optimize arbitrary designs of fin gripper bypassing time-consuming manufacturing. For example, the influence on the grasp forces of the passive torques, represented by the stiffness of the joints, can be evaluated to provide insights about practical designs.

Using the same 3-layer FRE finger model presented in Fig. 4 to validate the mathematical model with Adams, the influence of the stiffness can then be evaluated with respect to its location in the different parts of the finger. As a general rule for robotic fingers, it is desirable that the contact forces are positive to achieve a satisfactory envelopment and retention of the object. However, this is not always the case with underactuated or self-adaptive fingers as shown in Birglen et al. (2008); Birglen (2015). Hence, the percentage of positiveness of the contact forces provides a first performance metric to evaluate a robotic finger. This metric can be defined by:

$$\mu = \frac{\int_W \delta(\boldsymbol{\theta}) d\boldsymbol{\theta}}{\int_W d\boldsymbol{\theta}}, \tag{21}$$

where W is the workspace of the finger in terms joint angle range, while $\delta(\boldsymbol{\theta})$ is the Kronecker symbol for the positiveness of the contact forces, i.e.,

$$\delta(\boldsymbol{\theta}) = \begin{cases} 1 \text{ if all } f_i \geq 0 \text{ for } i = 1, 2, 3, \\ 0 \text{ otherwise.} \end{cases} \tag{22}$$

The value of μ is function of the joint angles $\boldsymbol{\theta}$ chosen as generalized coordinates. A second metric of interest is the average value of the coefficient of variation proposed by Birglen (2015) of the contact forces. This coefficient is defined as the ratio between the standard deviation and the average of the contact forces, namely:

$$c_v = \frac{\int_W (s/\bar{f}) \delta(\boldsymbol{\theta}) d\boldsymbol{\theta}}{\int_W \delta(\boldsymbol{\theta}) d\boldsymbol{\theta}} \tag{23}$$

where s is the standard deviation of the contact forces in a defined configuration and $\bar{f}$ is the average value of these forces. This coefficient of variation quantifies how balanced the contact forces are and is used to avoid situations where the forces are positive but very different one from another.

Additionally, the ratio between the sum of the contact forces and the actuation force at the base of the finger is also defined here since they both directly impact the grasp strength and the slip resistance and defined in Falco et al. (2015) as briefly mentioned in Sec. 4:

$$\eta_a^g = \frac{\int_W (f_\mathrm{g}/f_a)\delta(\boldsymbol{\theta})d\boldsymbol{\theta}}{\int_W \delta(\boldsymbol{\theta})d\boldsymbol{\theta}}, \tag{24}$$

where f_g is the summation of the magnitudes of all the contact forces in a defined configuration. Similarly, the ratio between the resulting forces applied on the object along the x- and y-axes is defined as:

$$\eta_y^x = \frac{\int_W (f_x/f_y)\delta(\boldsymbol{\theta})d\boldsymbol{\theta}}{\int_W \delta(\boldsymbol{\theta})d\boldsymbol{\theta}}. \tag{25}$$

All the previous metrics use a joint angle workspace defined by:

$$W \equiv \begin{cases} \pi/6 \leq \theta_1^F \leq \pi/2, \ i = 1 \\ 0 \leq \theta_i^F \leq \pi/3, \ i = 2, 3 \end{cases}, \tag{26}$$

To study the impact of the stiffness at the different locations in a FRE finger, the mathematical model of this paper was used by selectively activating (or not) the stiffnesses of the different joints. Namely, all architectures in the following section is designated by a triplet of binary digit (either 1 or 0). The first digit indicates if a unitary stiffness is assumed in the front beam joints while a zero points to the absence thereof, i.e. a joint with negligible stiffness in practice. The second digit is used for the back beam and the last one for the crossbeam joints. For example, "100" points to a design where $K_i^F = 1$, $K_i^B = 0$, and $K_j^{C\mathrm{f}} = K_j^{C\mathrm{b}} = 0$ for $i = 1, 2, 3$, and $j = 1, 2$ (note that $K_3^{C\mathrm{f}} = K_3^{C\mathrm{b}} \equiv 1$ by convention since this joint belongs to both the front and back beams). The results with the metrics previously discussed are listed in Table 9.

From Table 9, one can find that the positiveness of the contact forces appears to be more sensitive to the stiffness at the crossbeam joints. This result appears reasonable since the angles of rotation at the crossbeam joints typically vary considerably more than these of the beam joints in a certain configuration. Therefore, the torques produced by these joints is greater than at the front and back beams. It can also be observed that the back beam stiffness contributes to the percentage of the positive contact forces only marginally. In addition, a less obvious trend can be identified where the stiffness values at the front beam joints tend to lower the percentage μ. This last finding intuitively agrees though with the intuition that a stiffer front beam decreases the capability of the finger to adapt to the shapes of the objects, in which cases of negative contact forces occur at the phalanges that are not in contact. Comparing the grasp-to-actuation ratio η_a^g and the mean value of the contact-force sum $\bar{f}_\mathrm{g}$ for different stiffness settings, one can conclude that η_a^g does not fluctuate much, even when the contact-force sum is more than doubled. It should also be noticed that the $\bar{f}_y$ are all positive considering the sign of η_y^x, which means that the contact forces tend to push the grasped object upward while compressing it. Hence, sufficient friction appears to be needed at the interface of this FRE gripper to ensure a secure grasp. One can remark that this comment seems to correlate with the technical specifications of Festo's fingers that report a rather weak resistance to pullout forces (less than 25 N for an actuation force over 150 N) as one can see in Festo Gmbh (2017). In summary for this analysis, the set "011" generates the highest percentage of positive contact forces with a relatively low front-beam stiffness, while the set "010" generates the lowest variation of the contact forces with a relatively high crossbeam stiffness. The crossbeam stiffnesses therefore appear to be generally the most beneficial. The "010" design also achieve the largest x-to-y force ratio. These results suggest that practical designs should minimize the back and front stiffnesses compared to the crossbeam joints' and therefore should target a front and back beams as thin as possible. This was demonstrated experimentally using the 3D-printed fingers used in this paper by applying a rubber compound to locally reinforce and stiffen the crossbeam hinges (while keeping the front and back beams unchanged) which resulted in a significant increase of the contact forces (on average the compression force increased 32% experimentally) with little impact on the shape adaptation capability as discussed in a video summarizing this paper and available at: https://www.youtube.com/watch?v=MCh-CLse0as.

Table 9. Performance comparison with respect to the location of the passive torques

K_i^F, K_i^B, K_j^C	100	110	010	101	111	001	011
μ (%)	0.64	1.09	1.34	1.47	1.83	2.03	2.33
c_v	0.87	0.84	0.67	0.87	0.85	0.72	0.73
$\bar{f}_\mathrm{g}$ (mN)	13.1	46.2	15.5	26.2	32.9	23.2	29.8
η_a^g	1.23	1.26	1.24	1.17	1.16	1.19	1.17
η_y^x	2.42	2.38	6.24	2.75	3.10	4.05	4.32

Note: K_j^C stands for both $K_j^{C\mathrm{f}}$ and $K_j^{C\mathrm{b}}$

7 Conclusion

This paper presented a step-by-step analysis of a general FRE finger with passive elements in all its pseudo-joints. First, a pseudo-rigid-body model of this type of soft finger was constructed and its parameters identified either from geometry or FEA. This mathematical model was also validated by multibody simulations and supported by actual experimental measurements. In the last part of the paper, the usefulness of the model was shown in an example where the impact on the quality of the grasp from the stiffnesses in different locations of the finger was discussed. This opens the door toward the optimization of these FRE fingers for the task at hand.

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A Detailed Expressions for Equation (10)

$$\boldsymbol{A}_i=\begin{bmatrix} \boldsymbol{E\nu}_i^B+\boldsymbol{E\nu}_i^C & \boldsymbol{0}_{2\times1} & \boldsymbol{E\nu}_i^C \\ 1 & -1 & 1 \end{bmatrix},\quad \boldsymbol{E}=\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

$$\boldsymbol{V}_i=\begin{cases} \begin{bmatrix} \boldsymbol{E\nu}_1^F \\ 1 \end{bmatrix}, & i=1 \\ \begin{bmatrix} \boldsymbol{E}\left(\boldsymbol{\nu}_{i-1}^F+\boldsymbol{\nu}_i^F\right) & -\boldsymbol{E}\left(\boldsymbol{\nu}_{i-1}^B+\boldsymbol{\nu}_i^B+\boldsymbol{\nu}_i^C\right) & \boldsymbol{E\nu}_i^F \\ 1 & -1 & 1 \end{bmatrix}, & i>1 \end{cases}$$

$$\boldsymbol{\nu}_i^F=\frac{l_i^F}{2}\boldsymbol{u}_i^F+\boldsymbol{\varepsilon}_i^F,\quad \boldsymbol{\nu}_{i-1}^F=\frac{l_{i-1}^F}{2}\boldsymbol{u}_{i-1}^F-\boldsymbol{\varepsilon}_{i-1}^F,$$

$$\boldsymbol{\nu}_i^B=\frac{l_i^B}{2}\boldsymbol{u}_i^B+\boldsymbol{\varepsilon}_i^B,\quad \boldsymbol{\nu}_{i-1}^B=\frac{l_{i-1}^B}{2}\boldsymbol{u}_{i-1}^B-\boldsymbol{\varepsilon}_{i-1}^B,\quad i=1,\ldots,n,$$

B Detailed Expressions for the Joint Angles

$$\theta_i^B=\begin{cases}\theta_{01}^B+\psi_r-\psi_0, & i=1\\ \theta_{0i}^B, & i=2,\ldots,n\end{cases},\quad \theta_{0i}^B=\beta_i-\sigma_i-\left(\sum_{k=1}^{i-1}\theta_k^B-\zeta_{i-1}\right),\quad i=1,\ldots,n,$$

$$\beta_i=\delta_i+\arccos\left(\frac{-C_i}{\sqrt{A_i^2+B_i^2}}\right),\quad \delta_i=\arctan(B_i/A_i),$$

$$A_i=2v_i^Bv_{i-1}^{\mathrm{v}}-2v_i^Fv_i^B\cos\theta_i^{\mathrm{v}},\quad B_i=-2v_i^Fv_i^B\sin\theta_i^{\mathrm{v}},$$

$$C_i=(v_i^F)^2+(v_i^B)^2+(\iota_{i-1}^{\mathrm{v}})^2-(l_i^C)^2-2v_i^F\iota_{i-1}^{\mathrm{v}}\cos\theta_i^{\mathrm{v}},$$

$$\sigma_i=\arccos\left(\frac{\boldsymbol{u}_i^B\cdot\boldsymbol{v}_i^B}{|\boldsymbol{u}_i^B|\cdot|\boldsymbol{v}_i^B|}\right),\quad \zeta_{i-1}=\arccos\left(\frac{\boldsymbol{\iota}_{i-1}^{\mathrm{v}}\cdot\boldsymbol{x}}{\left|\boldsymbol{\iota}_{i-1}^{\mathrm{v}}\right|\cdot|\boldsymbol{x}|}\right),$$

$$\theta_i^{\mathrm{v}}=\theta_i^F-\sigma_i+\sum_{k=1}^{i-1}\theta_k^F-\psi_r+\psi_0-\zeta_{i-1},$$

$$\boldsymbol{\iota}_{i-1}^{\mathrm{v}}=\boldsymbol{\iota}_{i-2}^{\mathrm{v}}+l_{i-1}^B\left[\cos\left(\sum_{k=1}^{i-1}t_k^B\right)\sin\left(\sum_{k=1}^{i-1}t_k^B\right)\right]^T-l_{i-1}^F\left[\cos\left(\sum_{k=1}^{i-1}t_k^F\right)\sin\left(\sum_{k=1}^{i-1}t_k^F\right)\right]^T.$$

$$\theta_i^{\mathrm{fl}}=\gamma_i-(\pi-\psi_r+\theta_i^{\mathrm{v}})-(\psi_r-\psi_0),\theta_i^{\mathrm{fr}}\quad=\pi-(\psi_r-\zeta_i)-\lambda_i+\psi_r-\psi_0,$$

$$\gamma_i=\pi-\theta_i^{\mathrm{v}}+\phi_i,\ \lambda_i=\pi-\gamma_i-\theta_i^{\mathrm{v}}+\beta_i,\ \phi_i=\arctan\left(\frac{v_i^B\sin\beta_i-v_i^F\sin\theta_i^{\mathrm{v}}}{\iota_{i-1}^{\mathrm{v}}+v_i^B\cos\beta_i-v_i^F\cos\theta_i^{\mathrm{v}}}\right).$$

C Detailed Expressions for Equation (20)

$$\boldsymbol{A}_1=\begin{bmatrix} \frac{l_1^T}{2}\boldsymbol{E}\begin{bmatrix}\cos\theta_1^T\\ \sin\theta_1^T\end{bmatrix}+\varepsilon_1\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^T+\psi_0)\\ \sin(\theta_1^T+\psi_0)\end{bmatrix}+\boldsymbol{Eu}_1 & \boldsymbol{0}_2 & \boldsymbol{Eu}_1 \\ 1 & -1 & 1 \end{bmatrix},$$

$$\boldsymbol{u}_1=\frac{l_1^F}{2}\begin{bmatrix}\cos\theta_1^F\\ \sin\theta_1^F\end{bmatrix}+\varepsilon_1\begin{bmatrix}\cos(\theta_1^F-\psi_0)\\ \sin(\theta_1^F-\psi_0)\end{bmatrix}-\left(\boldsymbol{\iota}_0^{\mathrm{v}}+\frac{l_1^T}{2}\begin{bmatrix}\cos\theta_1^T\\ \sin\theta_1^T\end{bmatrix}+\varepsilon_1\begin{bmatrix}\cos(\theta_1^T+\psi_0)\\ \sin(\theta_1^T+\psi_0)\end{bmatrix}\right),$$

$$\boldsymbol{V}_1=\frac{l_1^F}{2}\boldsymbol{E}\begin{bmatrix}\cos\theta_1^F\\ \sin\theta_1^F\end{bmatrix}+\varepsilon_1\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F-\psi_0)\\ \sin(\theta_1^F-\psi_0)\end{bmatrix},$$

$$\boldsymbol{A}_2=\begin{bmatrix} \frac{l_2^T}{2}\boldsymbol{Ea}_2+\varepsilon_2\boldsymbol{Ea}_2'+\boldsymbol{Eu}_2 & \boldsymbol{0}_2 & \boldsymbol{Eu}_2 \\ 1 & -1 & 1 \end{bmatrix},$$

$$\boldsymbol{a}_2=\begin{bmatrix}\cos(\theta_1^T+\theta_2^T)\\ \sin(\theta_1^T+\theta_2^T)\end{bmatrix},\quad \boldsymbol{a}_2'=\begin{bmatrix}\cos(\theta_1^T+\theta_2^T+\psi_0)\\ \sin(\theta_1^T+\theta_2^T+\psi_0)\end{bmatrix}$$

$$\boldsymbol{u}_2=l_1^F\begin{bmatrix}\cos\theta_1^F\\ \sin\theta_1^F\end{bmatrix}+\frac{l_2^F}{2}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F)\\ \sin(\theta_1^F+\theta_2^F)\end{bmatrix}+\varepsilon_2\begin{bmatrix}\cos(\theta_1^F+\theta_2^F-\psi_0)\\ \sin(\theta_1^F+\theta_2^F-\psi_0)\end{bmatrix}-\left(\boldsymbol{\iota}_0^{\mathrm{v}}+l_1^T\begin{bmatrix}\cos\theta_1^T\\ \sin\theta_1^T\end{bmatrix}+\frac{l_2^T}{2}\begin{bmatrix}\cos(\theta_1^T+\theta_2^T)\\ \sin(\theta_1^T+\theta_2^T)\end{bmatrix}+\varepsilon_2\begin{bmatrix}\cos(\theta_1^T+\theta_2^T+\psi_0)\\ \sin(\theta_1^T+\theta_2^T+\psi_0)\end{bmatrix}\right),$$

$$\boldsymbol{V}_2=\begin{bmatrix}\boldsymbol{v}_{211} & \boldsymbol{v}_{212} & \boldsymbol{v}_{213}\\ 1 & -1 & 1\end{bmatrix},$$

$$\boldsymbol{v}_{211}=\frac{l_1^F}{2}\boldsymbol{E}\begin{bmatrix}\cos\theta_1^F\\ \sin\theta_1^F\end{bmatrix}-\varepsilon_1\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F-\psi_0)\\ \sin(\theta_1^F-\psi_0)\end{bmatrix}+\frac{l_2^F}{2}\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F)\\ \sin(\theta_1^F+\theta_2^F)\end{bmatrix}+\varepsilon_2\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F-\psi_0)\\ \sin(\theta_1^F+\theta_2^F-\psi_0)\end{bmatrix},$$

$$\boldsymbol{v}_{212}=-\frac{l_1^T}{2}\boldsymbol{E}\begin{bmatrix}\cos\theta_1^T\\ \sin\theta_1^T\end{bmatrix}-\varepsilon_1\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^T+\psi_0)\\ \sin(\theta_1^T+\psi_0)\end{bmatrix}+\frac{l_2^T}{2}\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^T+\theta_2^T)\\ \sin(\theta_1^T+\theta_2^T)\end{bmatrix}+\varepsilon_2\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^T+\theta_2^T+\psi_0)\\ \sin(\theta_1^T+\theta_2^T+\psi_0)\end{bmatrix}+\boldsymbol{Eu}_2,$$

$$\boldsymbol{v}_{213}=\frac{l_2^F}{2}\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F)\\ \sin(\theta_1^F+\theta_2^F)\end{bmatrix}+\varepsilon_2\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F-\psi_0)\\ \sin(\theta_1^F+\theta_2^F-\psi_0)\end{bmatrix},$$

$$\boldsymbol{A}_3=\begin{bmatrix} \frac{l_3^T}{2}\boldsymbol{Ea}_3+\varepsilon_3\boldsymbol{Ea}_3'+\boldsymbol{Eu}_3 & \boldsymbol{0}_2 & \boldsymbol{Eu}_3 \\ 1 & -1 & 1 \end{bmatrix},$$

$$\boldsymbol{a}_3=\begin{bmatrix}\cos\sum_{k=1}^{3}\theta_k^T\\ \sin\sum_{k=1}^{3}\theta_k^T\end{bmatrix},\quad \boldsymbol{a}_3'=\begin{bmatrix}\cos\left(\sum_{k=1}^{3}\theta_k^T+\psi_0\right)\\ \sin\left(\sum_{k=1}^{3}\theta_k^T+\psi_0\right)\end{bmatrix}$$

$$\boldsymbol{u}_3=l_1^F\begin{bmatrix}\cos\theta_1^F\\ \sin\theta_1^F\end{bmatrix}+l_2^F\begin{bmatrix}\cos(\theta_1^F+\theta_2^F)\\ \sin(\theta_1^F+\theta_2^F)\end{bmatrix}+\frac{l_3^F}{2}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F+\theta_3^F)\\ \sin(\theta_1^F+\theta_2^F+\theta_3^F)\end{bmatrix}+\varepsilon_3\begin{bmatrix}\cos(\theta_1^F+\theta_2^F+\theta_3^F-\psi_0)\\ \sin(\theta_1^F+\theta_2^F+\theta_3^F-\psi_0)\end{bmatrix}-\left(\boldsymbol{\iota}_0^{\mathrm{v}}+l_1^T\begin{bmatrix}\cos\theta_1^T\\ \sin\theta_1^T\end{bmatrix}+l_2^T\begin{bmatrix}\cos(\theta_1^T+\theta_2^T)\\ \sin(\theta_1^T+\theta_2^T)\end{bmatrix}+\frac{l_3^T}{2}\begin{bmatrix}\cos(\theta_1^T+\theta_2^T+\theta_3^T)\\ \sin(\theta_1^T+\theta_2^T+\theta_3^T)\end{bmatrix}+\varepsilon_3\begin{bmatrix}\cos(\theta_1^T+\theta_2^T+\theta_3^T+\psi_0)\\ \sin(\theta_1^T+\theta_2^T+\theta_3^T+\psi_0)\end{bmatrix}\right),$$

$$\boldsymbol{V}_3=\begin{bmatrix}\boldsymbol{v}_{311} & \boldsymbol{v}_{312} & \boldsymbol{v}_{313}\\ 1 & -1 & 1\end{bmatrix},$$

$$\boldsymbol{v}_{311}=\frac{l_2^F}{2}\boldsymbol{E}\begin{bmatrix}\cos\theta_1^F+\theta_2^F\\ \sin\theta_1^F+\theta_2^F\end{bmatrix}-\varepsilon_2\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F-\psi_0)\\ \sin(\theta_1^F+\theta_2^F-\psi_0)\end{bmatrix}+\frac{l_3^F}{2}\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F+\theta_3^F)\\ \sin(\theta_1^F+\theta_2^F+\theta_3^F)\end{bmatrix}+\varepsilon_3\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F+\theta_3^F-\psi_0)\\ \sin(\theta_1^F+\theta_2^F+\theta_3^F-\psi_0)\end{bmatrix},$$

$$\boldsymbol{v}_{312}=-\frac{l_2^T}{2}\boldsymbol{E}\begin{bmatrix}\cos\theta_1^T+\theta_2^T\\ \sin\theta_1^T+\theta_2^T\end{bmatrix}-\varepsilon_2\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^T+\theta_2^T+\psi_0)\\ \sin(\theta_1^T+\theta_2^T+\psi_0)\end{bmatrix}+\frac{l_3^T}{2}\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^T+\theta_2^T+\theta_3^T)\\ \sin(\theta_1^T+\theta_2^T+\theta_3^T)\end{bmatrix}+\varepsilon_3\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^T+\theta_2^T+\theta_3^T+\psi_0)\\ \sin(\theta_1^T+\theta_2^T+\theta_3^T+\psi_0)\end{bmatrix}+\boldsymbol{Eu}_3,$$

$$\boldsymbol{v}_{313}=\frac{l_3^F}{2}\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F+\theta_3^F)\\ \sin(\theta_1^F+\theta_2^F+\theta_3^F)\end{bmatrix}+\varepsilon_3\boldsymbol{E}\begin{bmatrix}\cos(\theta_1^F+\theta_2^F+\theta_3^F-\psi_0)\\ \sin(\theta_1^F+\theta_2^F+\theta_3^F-\psi_0)\end{bmatrix},$$

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